



Potential of sodium dodecyl sulfate micellar solutions as eluents in magnetic dispersive micro-solid phase extraction with polydopamine-coated magnetite nanoparticles. Application to antidepressant drugs

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ABSTRACT

In this paper, the potential of micellar solutions of the anionic surfactant sodium dodecyl sulfate (SDS) as eluents in dispersive micro-solid phase extraction (D- μ SPE) using polydopamine-coated magnetite nanoparticles (Fe_3O_4 @PDA NPs) for the extraction and preconcentration of seven basic drugs (bupropion, citalopram, fluoxetine, mianserin, nomifensine, trimipramine, and viloxazine) is explored for the first time (to the best of our knowledge) and compared with conventional hydro-organic eluents. The impact of the sample solution pH, Fe_3O_4 @PDA NPs and PDA coating amounts and extraction time on the extraction efficiency (EE), as well as the composition of the eluent on the overall efficiency (OE) are studied. Under the selected experimental conditions (50 mg of Fe_3O_4 @PDA NPs, 100 μL of 1 M NH_3 , 5 min of extraction time and 0.15 M SDS at pH 2.6 as eluent), EE and OE values were higher than 90% for all compounds and for the most hydrophobic compounds (trimipramine, fluoxetine and mianserin), respectively. The results shown in this paper demonstrate the suitability of Fe_3O_4 @PDA NPs as a sorbent for the extraction of antidepressants as well as the advantages of using SDS micellar solutions over classic hydro-organic eluents containing methanol, acetonitrile or tetrahydrofuran. Finally, the stability and reusability of the Fe_3O_4 @PDA NPs is proven.

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1. Introduction

Currently, dispersive micro-solid phase extraction (D- μ SPE) is gaining increasing interest because it is a simple, fast and cost-effective sample preparation technique that provides high preconcentration factors. In D- μ SPE, a wide variety of micro- and nanomaterials can be used as sorbents, which determine the selectivity and efficiency of the method [1–4]. The larger surface area-to-volume ratio of nanomaterials results in an improvement of the extraction process over micromaterials. In the literature, a broad diversity of nanosorbents have been used, such as metallic and metallic oxide nanoparticles (NPs), carbon and silicon nanomate-

rials, nanofibers, nanoclays, polymer-based nanoadsorbents, aerogels, etc. [5] for the extraction of different analytes in various complex matrices [1–4].

Magnetic NPs composed of a magnetic core of iron, nickel, cobalt or one of their oxides are of particular interest in D- μ SPE. In addition to the intrinsic advantages of nanomaterials, they present high magnetic moments, so they can be separated using an external magnetic field, avoiding filtration or centrifugation steps, and reducing the analysis time [6–8]. Magnetite (Fe_3O_4) NPs are of special importance due to their simple synthesis, biocompatibility, biodegradability and excellent magnetic properties [9]. The surface of the Fe_3O_4 NPs can be easily modified with inorganic or organic substances (e.g. silica, alumina, graphene, molecularly-imprinted polymers, surfactants, etc.) to improve the selectivity of the extraction [7,10].

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Among the different coatings currently proposed, polydopamine (PDA) coatings are particularly noteworthy. PDA, obtained by oxidative polymerization of dopamine, is a mixture of oligomers that differ in the number and composition of monomers and can form supramolecular structures [11–14]. PDA exhibits strong adhesion to all types of organic substrates and metal ions due to its structure comprising a wide variety of functional groups (amino, imine, phenyl, phenol, etc.), its amphipathic character [15] and the presence of a highly delocalized π - π conjugated system [14].

Polydopamine-coated magnetite nanoparticles (Fe_3O_4 @PDA NPs) have recently aroused great interest in magnetic D- μ SPE because of their easy preparation, good dispersibility in water, biodegradability and high potential for the extraction of analytes of different nature.

These NPs have been used for the extraction of polycyclic aromatic hydrocarbons [16], benzopyrene [17], endocrine disrupting compounds and pharmaceuticals [18] and phenolic compounds [19] from waters; estrogens from waters [20–22] and milks [23]; polyphenolic antioxidants [24] and berberine [25] from plant extracts; aflatoxins from red wine [26]; synthetic colorants from beverages [15]; and copper from foods [27]. In these studies, the extraction efficiency values vary between 55–112%. Expanding the applicability to new analytes and matrices will further confirm the high potential of these materials as sorbents in D- μ SPE.

The success of a D- μ SPE procedure depends on the efficiency of the extraction and the elution steps in which electrostatic, hydrophobic and/or steric interactions between the solute and both the sorbent and eluent solution are involved. To obtain adequate efficiency values, the experimental conditions (e.g., pH and polarity of the medium) for the elution process should be opposite to those used in the extraction process to minimize analyte-sorbent affinity.

For compounds present in aqueous matrices, the pH of the sample is usually adjusted to favor the extraction and organic (usually, pure acetonitrile and methanol) or hydro-organic solutions are used for the elution of compounds. However, in some cases, although the extraction efficiency values are adequate, it is difficult to elute the compounds, which results in low efficiency values for the overall process.

Surfactants are amphiphilic compounds, which contain a non-polar hydrocarbon-chain bonded to a polar group (anionic, cationic, neutral or zwitterionic). At concentrations over the critical micellar concentration (cmc), the surfactant monomers tend to aggregate to form micelles with spherical structure. In aqueous solutions, the hydrophobic chain of the surfactant is located in the core, whereas the hydrophilic or charged heads are oriented to the water. Therefore, micelles have non-polar and polar (and electrostatic for ionic surfactants) solubilization sites for compounds with a wide range of hydrophobicity, charge and molecular mass [28]. To our knowledge, micellar solutions have not been used as eluents in D- μ SPE.

The studies included in this paper constitute a first attempt on the capabilities of micellar solutions as eluents in D- μ SPE. As an example, in this paper the potential of micellar solutions of the anionic surfactant sodium dodecyl sulfate (SDS) as eluents in D- μ SPE using Fe_3O_4 @PDA NPs for the extraction and preconcentration of seven basic drugs (bupropion, citalopram, fluoxetine, mianserin, nomifensine, trimipramine, and viloxazine) is explored and compared with conventional hydro-organic solvents. A study of the parameters which affect the efficiency of the extraction and elution steps is performed, and some strategies are recommended. To the best of our knowledge, this is the first time that Fe_3O_4 @PDA NPs are used as sorbent for antidepressant extraction.

The studies shown in this work constitute a starting point for the further development of D- μ SPE analytical procedures for the determination of these compounds in different matrices.

2. Experimental

2.1. Instrumentation

An Agilent Technologies 1100 chromatograph (Palo Alto, CA, USA) with an isocratic pump, a UV-visible variable wavelength detector, a column thermostat and an autosampler was used. Data acquisition and processing were performed by means of the OpenLAB CDS ChemStation for LC&LC/KS systems (A.02.13[04], ©Agilent Technologies 2001–2006).

Mobile phase solutions were vacuum-filtered through 0.45 μm Nylon membranes (Micron Separations, Westboro, MA, USA) and degassed in an Elmasonic S60 ultrasonic bath (Elma, Singen, Germany). Ultra-Clear TWF UV deionized water (SG Water, Barsbüttel, Germany) was used to prepare the solutions. A Crison MicroPH 2000 pHmeter (Crison Instruments, Barcelona, Spain) was employed to adjust the pH of the buffers.

Before injection into the chromatographic system, solutions were filtered using disposable 0.22 μm Nylon syringe filters (Análisis Vínicos, S.L., Tomelloso, Spain).

For the D- μ SPE procedure, 25 mL screw cap glass vials from Deltalab (Barcelona, Spain), an overhead shaker from Heidolph, (Walpersdorfer, Germany) and a neodymium, N48, nickel-plated magnet (50×15×15 mm) from supermagnete (Webcraft GmbH, Göttingen, Germany) were used.

For the characterization of Fe_3O_4 NPs, X-ray powder diffraction (XRD) was performed by the Servicio Central de Soporte a la Investigación Experimental (SCSIE) of the University of Valencia using a Bruker D8 Advance A25 diffractometer (Bruker AXS GmbH, Karlsruhe, Germany) with monochromatic Cu K α source operated at 40 kV and 40 mA. Patterns were recorded over the angular range 10 – 80° (2 θ). For the evaluation of the microstructure, size and morphology of the Fe_3O_4 and Fe_3O_4 @PDA NPs, scanning electron microscopy (SEM) images were acquired on a field emission scanning electron microscope Hitachi S4800 (Hitachi High-Tech Corporation, Tokyo, Japan) operating at 20 kV. To estimate the PDA coating of the Fe_3O_4 @PDA NPs, thermal gravimetric analysis (TGA) curves were obtained using a Setaram Setsys 16/18 thermobalance (Setaram Instrumentation, Caluire-et-Cuire, France) under an air atmosphere, using a flow rate of 25 mL·min⁻¹ and heating rate of 5 °C·min⁻¹. XRD, SEM and TGA results are shown in Supplementary materials.

Zeta potential measurements of Fe_3O_4 @PDA NPs were carried out using a Zetasizer Nano-ZS from Malvern Panalytical Ltd. (Spectris company, London, UK). Zeta potential experiments were performed using water as dispersant (refraction index: 1.330; viscosity: 0.8872×10⁻³ Pa·s; dispersant dielectric constant: 78.5) at 25°C.

2.2. Chemicals and solutions

Bupropion hydrochloride and mianserin hydrochloride were from Alfa Aesar (Thermo Fisher Scientific Inc., Karlsruhe, Germany); citalopram hydrobromide was from Tokyo Chemical Industry (Tokyo, Japan); nomifensine maleate was from Sigma-Aldrich (©Merck KGaA, Darmstadt, Germany) and trimipramine maleate was from Cayman Chemical Co (Ann Arbor, MI, USA). Fluoxetine hydrochloride was kindly donated by Alter (Madrid, Spain), and viloxazine hydrochloride by Astra Zeneca (Cheshire, UK).

Drug stock standard solutions of 100 mg·L⁻¹ were prepared by dissolving the adequate amount of each compound in methanol. For the D- μ SPE experiments, working solutions (prepared by dilution in water of the appropriate volume of the stock standard solution) of 1 mg·L⁻¹ were used for bupropion, and mianserin; and 2 mg·L⁻¹ for citalopram, trimipramine, fluoxetine, nomifensine and viloxazine as samples.

All reagents were of analytical grade. Ammonium acetate, ammonia 32%, tris(hydroxymethyl)aminomethane (Tris), tetrahydrofuran (THF), 1-propanol (1-PrOH) and 2-propanol (2-PrOH) were from Scharlau S.L. (Barcelona, Spain). Acetonitrile (ACN) and methanol (MeOH) (HPLC grade) were from PanReac AppliChem (Barcelona, Spain). Sodium dodecyl sulfate (SDS) 99% was from Acros Organics (Geel, Belgium). Hydrochloric acid 37%, and formic acid 99% were from VWR Chemicals (Radnor, USA). $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, and dopamine hydrochloride were from Sigma-Aldrich.

10 mM ammonium/ammonia buffer at pH 8.0 was prepared by dissolving the appropriate amount of ammonium acetate in water and adjusting potentiometrically the pH with ammonia 2.5 M. Mobile phases were prepared by mixing 10 mM ammonium acetate (pH 8.0) with ACN to obtain the working concentration.

10 mM formic acid/formate buffer (FB, pH 2.6) was prepared by dissolving the appropriate amount of ammonium formate in water and adjusting potentiometrically the pH with hydrochloric acid 2.5 M.

10 mM Tris buffer (pH 8.5) was prepared by dissolving the appropriate amount of Tris in water and adjusting potentiometrically the pH with hydrochloric acid 2.5 M.

Micellar solutions of SDS at pH 2.6 were prepared by dissolving the appropriate amount of SDS in 10 mM FB (pH 2.6) to give a final surfactant concentration of 0.05, 0.15 and 0.30 M.

2.3. Synthesis and characterization of Fe_3O_4 @PDA NPs

Fe_3O_4 @PDA NPs were synthesized and characterized at the laboratories of the Materials Science Institute of the University of Valencia (ICMUV) following the procedure described in [29,30]. Briefly, Fe_3O_4 NPs were prepared by the chemical coprecipitation method. In a three-necked flask under N_2 atmosphere, 3.0 g of $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$ and 6.1 g of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ were mixed and dissolved in 20 mL of water by means of an ultrasonic bath. This solution was heated at 80 °C and 20 mL of a $\text{H}_2\text{O}:\text{NH}_3$ (1:3, v:v) solution was added slowly. The obtained suspension was stirred at 100 °C for 2 h. Next, the suspension was cooled at room temperature and the solid, Fe_3O_4 NPs, was separated with a neodymium magnet, washed with a mixture of $\text{H}_2\text{O}:\text{MeOH}$ (50:50, v:v), and dried in an oven at 70 °C for 24 h. Then, ~ 600 mg of the synthesized Fe_3O_4 NPs were stirred in 60 mL of 10 mM Tris buffer (pH 8.5) in the presence of ~ 210 mg of dopamine hydrochloride during 24 h to form the layers of PDA on magnetite. Next, the Fe_3O_4 @PDA NPs were separated with a neodymium magnet, washed with a mixture of $\text{H}_2\text{O}:\text{MeOH}$ (50:50, v:v), and dried at 70 °C overnight. The Fe_3O_4 @PDA NPs were stored at room temperature.

2.4. D- μ SPE procedure

20 mL of the sample solution (analyte working solution) and 100 μL of 1 M ammonia were added to a 25 mL stoppered glass vial containing 50 mg of Fe_3O_4 @PDA NPs. The mixture was shaken for 5 min at 80 rpm in an overhead shaker. The vial was placed on a neodymium magnet for 30 s to allow the NPs to settle. The supernatant (residual solution, used for estimating the extraction efficiency) was decanted into a beaker, filtered and introduced into an HPLC vial for later injection into the chromatographic system. Elution of the analytes from Fe_3O_4 @PDA NPs were carried out in two steps with 500 μL (a total of 1 mL) of the eluent by sonication of the mixture for 5 min. After magnetic separation, the extracts (eluate, used for estimating the overall efficiency) were transferred and combined into an HPLC vial for chromatographic analysis. All experiments were performed in duplicate.

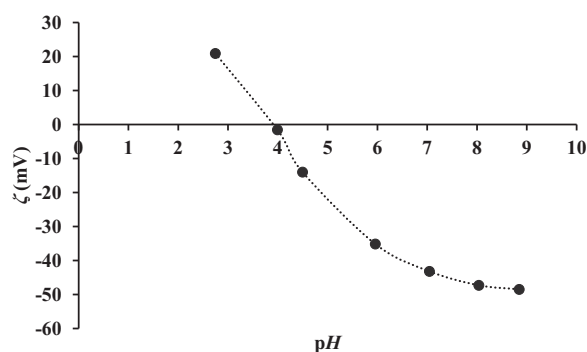


Fig. 1. Zeta potential (ζ) of Fe_3O_4 @PDA NPs at different pH values

2.5. HPLC analysis

The sample, residual and eluate solutions were analyzed by HPLC using a Halo C18 column (2.7 μm , 75 $\times$ 4.6 i.d. mm, Advanced Materials Technology Inc., Wilmington, Delaware, USA) and binary mixtures of 10 mM ammonium acetate buffer at pH 8.0 and ACN. The percentages (v:v) of ACN in the mobile phase were: 65% for the determination of mianserin and trimipramine; 55% for bupropion, citalopram and fluoxetine; 45% for nomifensine and 30% for viloxazine. The detection was performed in the UV at 220 nm. The mobile phase flow rate was 1.0 mL $\cdot$ min $^{-1}$, the injection volume was 5 μL and the temperature was 25 °C, in all experiences. Under these experimental conditions retention factors were lower than 4.

2.6. Software and calculations

Microsoft Excel (©Microsoft Corporation, version 2016) was used for calculation and data processing. Extraction efficiency (EE) and overall efficiency (OE) values were obtained using the following equations:

$$EE(\%) = 100 - \frac{A_{\text{RES}}}{A_{\text{P}_0}} \cdot 100 \quad (1)$$

$$OE(\%) = \frac{A_{\text{EL}}}{A_{\text{P}_0}} \cdot \frac{V_f}{V_0} \cdot 100 \quad (2)$$

where A_{RES} , A_{EL} , and A_{P_0} are the peak areas of the analyte obtained from the chromatographic analysis of the residual, eluate and sample solutions, respectively. V_f is the total volume of eluent (1 mL) and V_0 is the volume of the sample (20 mL).

3. Results and discussion

Electrostatic interactions analyte-NPs are key factors in the extraction and elution steps for ionizable compounds. PDA contains ionizable groups, so Fe_3O_4 @PDA NPs charge depends on the pH of the sample and the eluent solution. Fig. 1 shows the zeta potential (ζ) at different pH values, between 2.7 and 8.8, for the Fe_3O_4 @PDA NPs. As can be observed, the Fe_3O_4 @PDA NPs have a positive charge up to pH 4, isoelectric point, where a charge change occurs, becoming negative up to pH ~ 9.

Table 1S shows the chemical structures, the values of the minus logarithm of the acidity constants (pK_a), the logarithm of the *n*-octanol/water partition coefficient ($\log P$) and the values of the net charge at pH 2.5 and 8.5 of the compounds studied. As can be seen, the antidepressants studied present a variable hydrophobic character (i.e., $\log P$ values ranged between 1.23 and 4.51 for viloxazine and trimipramine, respectively). In addition, all of them, have only amino groups in their structure, so when ionized they present a positive net charge.

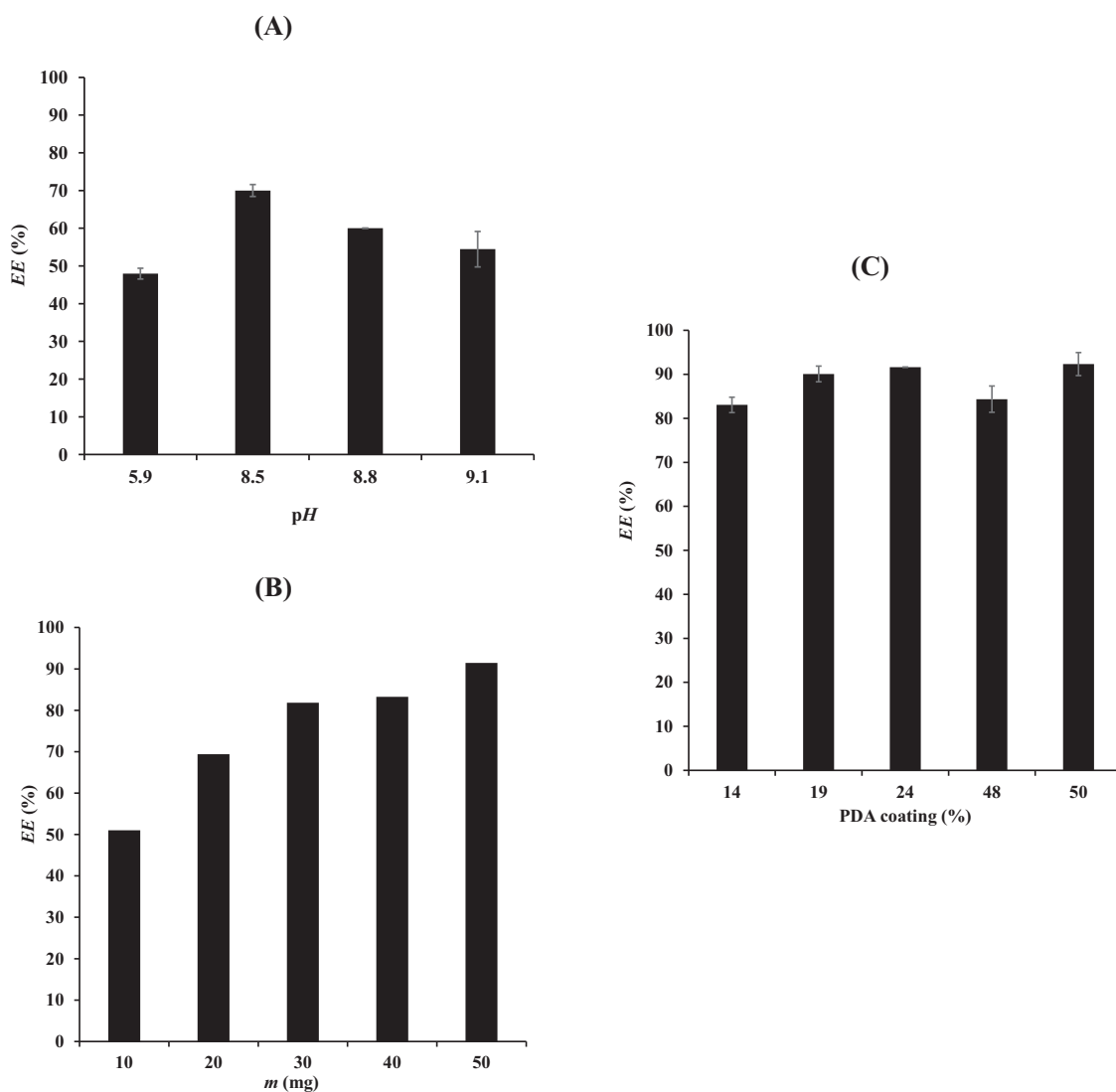


Fig. 2. Impact on extraction efficiency (EE) for bupropion of the following extraction conditions: (A) pH of sample solution, (B) mass of Fe_3O_4 @PDA NPs, and (C) percentage of PDA coating in the Fe_3O_4 @PDA NPs. Mean EE values together with their standard deviations are shown. See further experimental details in Section 2.4.

The following sections include several studies on the experimental variables which could affect the extraction (sample solution pH, Fe_3O_4 @PDA NPs and PDA coating amounts and extraction time) and elution (composition of the eluent) steps.

3.1. Impact of the extraction conditions on the extraction efficiency

The results reported in this section were obtained from the HPLC-UV analysis of the residual solutions collected after the extraction step (see Section 2.4) in the experimental conditions depicted in Section 2.5 and the subsequent estimation of EE values by means of Eq. 1 (see Section 2.6). The mobile phase composition (%ACN, v/v) was optimized for each compound according to its hydrophobicity (see Table S3).

For the initial study, bupropion was selected as model analyte due to its intermediate physicochemical characteristics with respect to the rest of the compounds. The following initial experimental extraction conditions were set (unless otherwise indicated): 20 mL of sample solution (1 mg·L⁻¹ bupropion), 100 μL of 1 M NH_3 , 50 mg of Fe_3O_4 @PDA NPs, and 10 min of extraction time. Fig. 2 shows the results obtained.

First, we studied the influence of the sample solution pH on the EE values. For this purpose, different volumes (0, 100, 250 and

500 μL) of 1 M NH_3 were added to the bupropion sample solution while the rest of variables were kept constant (in this study, 20 mg of Fe_3O_4 @PDA NPs were used). After the extraction step, the pH of the residual solutions was measured. As can be observed in Fig. 2A, the pH of the sample solution is a critical aspect in the extraction step. The increase of the solution pH up to 8.5 increased the EE values. From this pH value, the EE values slightly decreased. It must be taken into account that the highest Fe_3O_4 @PDA NPs negative charge is found at about pH > 8 (see Fig. 1), whereas the cationic form of bupropion predominates at pH values lower than 8.2 (see Table 1S) and the neutral form predominates at higher pH values. Thus, the lower EE value at pH 5.9 is due to lower Fe_3O_4 @PDA NPs negative charge at this pH, while the decreased in EE values from pH 8.5 could be attributed to the reduction in the net charge of bupropion (+0.34, +0.20 and 0.11, at pH 8.5, 8.8 and 9.1, respectively). This reduction of positive charge seems to be greater than the increase of the Fe_3O_4 @PDA NPs negative charge at these pH values, resulting in a weakening of electrostatic interactions. The results obtained indicate that, in addition to other types of interactions, electrostatic attraction plays a fundamental role in the extraction of bupropion with Fe_3O_4 @PDA NPs. In view of the results, 100 μL of 1 M NH_3 was added to the sample solution for further experiments.

Fig. 2.B shows the effect of the amount of $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs on the *EE* values for bupropion. As can be seen, *EE* values increased as the nanoparticles amount increased. The maximum extraction efficiency, (*EE* = 91%) was obtained using 50 mg of $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs. These results indicate that the use of NPs amounts below 50 mg did not allow the complete extraction of the compound, as the amount of bupropion in the solution exceeded the binding capacity of the $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs. Assuming that 50 mg of sorbent is the minimum amount necessary to provide adequate *EE*, the approximate binding capacity estimated from the data in Figure 2B is 0.4 μg bupropion/mg $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs. 50 mg of nanoparticles was selected for future experiments.

The impact of the percentage of PDA coating of the $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs on the *EE* of bupropion is shown in Fig. 2.C. Experiments were carried out using $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs coated with ~15–50% PDA. As can be observed, the *EE* values were similar (between 83 and 92%) for all the experiments; so, we did not consider this variable in the rest of experiences. Additionally, experiences were also carried out using Fe_3O_4 NPs not coated with polydopamine. Binding of bupropion to unmodified magnetite nanoparticles was observed. However, the *EE* values obtained were lower than those for magnetite nanoparticles coated with polydopamine, also showing a higher variability (results not shown).

Using the selected experimental conditions, the *EE* values were obtained for the rest of compounds. Table 2S shows the results obtained using extraction times of 5 or 10 min. For both extraction times, *EE* values were higher than 90% for all the compounds. It must be pointed out that even for mianserin the result was satisfactory (*EE* (%) = 95 ± 3), despite having a net charge close to zero at $\text{pH} = 8.5$ (see Table 1S). This result indicates that analyte- $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs interactions other than electrostatic ones may also be important. An extraction time of 5 min was used for further experiments.

3.2. Impact of the elution conditions on the overall efficiency

Using the above selected extraction conditions, we assayed different eluents following the procedure described in Section 2.4. For each eluent, the *OE* values were calculated using Eq. 2 (see Section 2.6).

As indicated in the introduction section, to obtain adequate *OE* values, the elution conditions should be opposite to the extraction conditions. Under the extraction conditions ($\text{pH} 8.5$), $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs are negatively charged, and the compounds studied have close-to-zero or positive charge. Therefore, in order to decrease the affinity of the analyte towards the NPs and favor it towards the eluent, several strategies are possible:

- The use of aqueous buffer solutions at pH values below 4 ($\text{Fe}_3\text{O}_4\text{@PDA}$ NPs and all the analytes present positive charge) to provoke analyte- $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs electrostatic repulsion.
- To increase the hydrophobic character of the elution solution by adding organic solvents (e.g., ACN, MeOH or THF) to aqueous buffer solutions at $\text{pH} < 4$ to favor analyte-sorbent electrostatic repulsion and analyte-eluent hydrophobic interactions.
- The use of micellar solutions buffered at $\text{pH} < 4$ that compete with $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs for the retention of the compounds, either by electrostatic attraction or by hydrophobic interaction.

At this stage, preliminary studies were also conducted using bupropion as model analyte. Non-micellar eluents assayed included 10 mM FB at $\text{pH} 2.6$ and binary mixtures (90:10, v:v) of several organic solvents (ACN, MeOH and THF) and FB (10 mM, $\text{pH} 2.6$). The use of THF:FB (10 mM, $\text{pH} 2.6$), 90:10 (v:v) solution as eluent led to a very dark brown suspension not suitable for injection into the chromatographic system. This fact could be attributed to the removal of the PDA coating. It must be also pointed out that

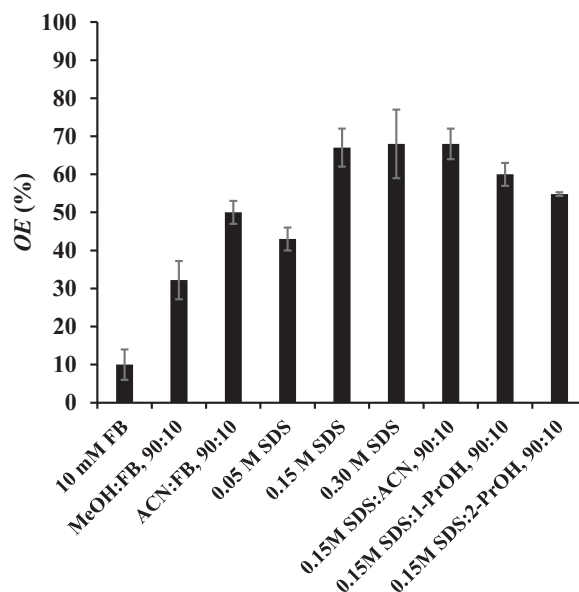


Fig. 3. Impact of the composition of the eluent on the overall efficiency (*OE*) for bupropion. Mean *OE* values together with their standard deviations are shown. See further experimental details in Section 2.4.

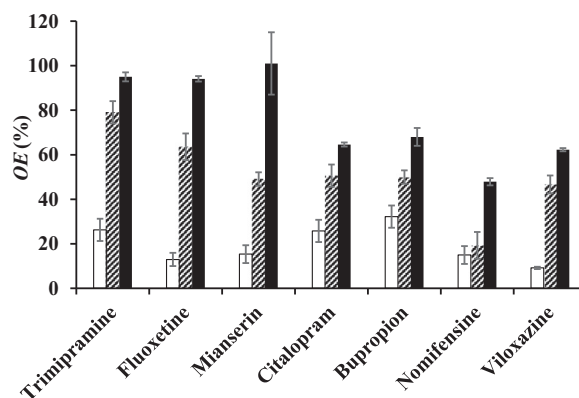


Fig. 4. Impact of the composition of the eluent on the overall efficiency (*OE*) for each compound studied using: (□) MeOH:FB ($\text{pH} = 2.6$) 90:10 (v:v); (▨) ACN:FB ($\text{pH} = 2.6$) 90:10 (v:v) and (■) SDS 0.15 M ($\text{pH} = 2.6$) as eluents. Compounds are shown in decreasing order of hydrophobicity. FB: 10 mM formic acid/formate buffer.

the use of ACN eluents also gave colored eluates, although they were suitable solutions for injection into the HPLC system. This fact has been also previously reported [20]. In addition, we observed that color intensity varied between experiments. In our opinion, these observations could be attributed to a partial and variable dissolution of PDA coating. This fact precludes the reusability of the $\text{Fe}_3\text{O}_4\text{@PDA}$ NPs.

As can be observed in Fig. 3, none of the non-micellar eluents assayed provided complete elution of bupropion. The worst result (*OE* (%) = 10 ± 4) was obtained with 10 mM FB at $\text{pH} 2.6$ and the best (*OE* (%) = 50 ± 3) with ACN:FB (10 mM, $\text{pH} 2.6$), 90:10 (v:v).

As stated in the introduction section, at our knowledge, micellar solutions have never been used as eluents in D- μSPE , despite their well-known high solubilizing power for a wide variety of compounds. To explore their capabilities, taking into account the positive charge of bupropion (as well as of all the other analytes) at pH values lower than 8.5, we opted for the anionic surfactant SDS. To this effect, 0.05, 0.15 and 0.30 M SDS micellar solutions at $\text{pH} 2.6$ were used as eluents. We found that the use of 0.15 and 0.30 M SDS solutions provided the best *OE* values (~70%, see Fig. 3).

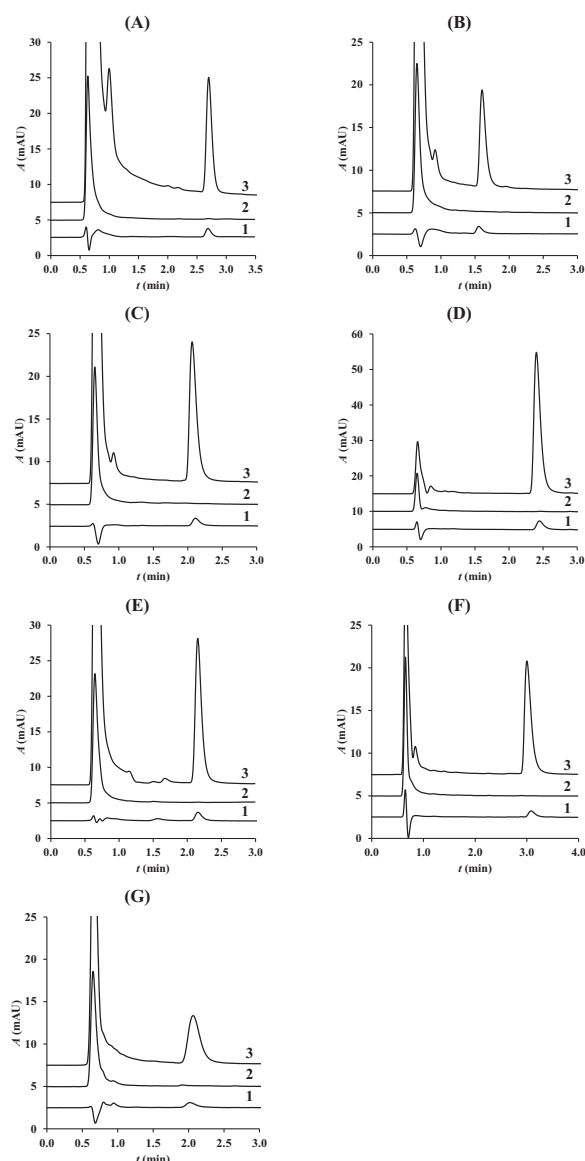


Fig. 5. Chromatograms corresponding to: (A) bupropion, (B) citalopram, (C) fluoxetine, (D) mianserin, (E) nomifensine, (F) trimipramine, and (G) viloxazine. (1) Sample (2) residual and (3) eluate solutions obtained using 0.15 M SDS (pH = 2.6) as eluent. See experimental details in Sections 2.4 and 2.5.

Moreover, the addition of a 10% (v:v) of ACN, 1-PrOH or 2-PrOH to the 0.15 M SDS (pH 2.6) solutions (commonly used as hybrid mobile phases in micellar liquid chromatography) did not improve the OE values.

For all the aforementioned reasons, we assayed the 0.15 M SDS (pH 2.6) micellar solution as eluent for the other analytes. We also tested MeOH:FB (10 mM, pH 2.6), 90:10 (v:v) and ACN:FB (10 mM, pH 2.6), 90:10 (v:v) eluents for comparative purposes. Fig. 4 shows the OE values obtained for each compound and each eluent used. As can be observed, for all the analytes, the best results were obtained using 0.15 M SDS (pH 2.6) micellar solution as eluent. The highest OE values using this eluent were obtained for mianserin, trimipramine and fluoxetine (OE (%) = 101 ± 14 , 95 ± 2 and 94.1 ± 1.2 , respectively), the most hydrophobic compounds. For the rest of compounds, OE values comprised between (47.9 ± 1.6) % for nomifensine and (68 ± 4) % for bupropion were obtained. On the other hand, the eluent containing ACN provided slightly lower OE values than the one consisting of SDS for all the compounds. Again,

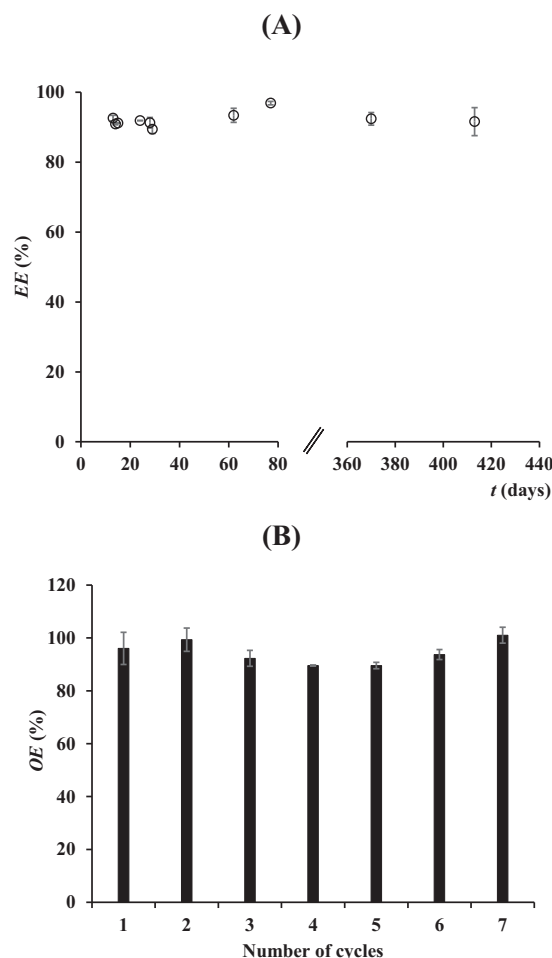


Fig. 6. Stability and reusability of $Fe_3O_4@PDA$ NPs. (A) Stability (along days) in terms of extraction efficiency (EE) for bupropion. (B) Reusability in terms of overall efficiency (OE) for mianserin. Extraction conditions: 50 mg of $Fe_3O_4@PDA$ NPs, 20 mL of the sample solution containing 1 mg/L of analyte, pH 8.5 (100 μ L of 1 M ammonia) and 5 min of extraction time. Elution conditions (Fig. B): $2 \times 500 \mu$ L of 0.15 M SDS at pH 2.6.

the aqueous ACN eluent provided eluates with variable color intensity, which limits the reusability of the $Fe_3O_4@PDA$ NPs.

Fig. 5 shows, for each compound, the chromatograms corresponding to the sample, the residual and eluate solutions under the extraction/elution conditions that provided the best results (50 mg of $Fe_3O_4@PDA$ NPs, 100 μ L of 1 M NH_3 , 5 min of extraction time and 0.15 M SDS at pH 2.6 as eluent; see further details in Section 2.4).

3.3. Stability and reusability of $Fe_3O_4@PDA$ NPs

The $Fe_3O_4@PDA$ NPs used in this work were synthesized one year ago and stored at room temperature. These NPs were used for approximately one-year period for experiments with bupropion. Fig. 6A shows the EE values obtained using these NPs during 420 days for experiments with bupropion. As can be observed, the EE values remained constant (EE (%) = 92 ± 2) during the entire period under study, which demonstrates the good stability of these NPs.

We also investigated the reusability of the $Fe_3O_4@PDA$ NPs. For this purpose, successive cycles of the d- μ SPE procedure described in Section 2.4 were performed using the same NPs for mianserin. After each cycle, the $Fe_3O_4@PDA$ NPs were washed with

a H₂O:MeOH (50:50, v:v) solution and dried at 70 °C overnight. Fig. 6.B shows the OE values obtained. As can be observed, no relevant differences were obtained in OE values when NPs were reused for at least 7 times, indicating a good reusability of the Fe₃O₄@PDA NPs. Only small variations between results obtained after washing cycles can be appreciated, which can be attributed to the reproducibility of the methodology applied. Good stability and reusability are key strengths of this sorbent.

4. Conclusions

In this paper, the ability of Fe₃O₄@PDA NPs and SDS micellar solutions to extract and elute seven basic drugs (bupropion, citalopram, fluoxetine, mianserin, nomifensine, trimipramine, and viloxazine) is proven for the first time.

The pH of the media is a key factor in both, the extraction and the elution steps, since the charge of Fe₃O₄@PDA NPs and the analytes depends on pH.

At pH 8.5, Fe₃O₄@PDA NPs has strong affinity towards the antidepressants studied (*EE* > 90%) due to the antidepressant-Fe₃O₄@PDA NPs electrostatic attraction; although other types of interactions cannot be neglected.

At pH 2.6, antidepressant-Fe₃O₄@PDA NPs electrostatic repulsion exists, but it is not strong enough to elute the analytes. In fact, the use of MeOH:FB (10 mM, pH 2.6), 90:10 (v:v) solutions as eluents provided very poor results (*OE* < 30% for all compounds). The use of eluents containing THF or ACN provoked the removal of the PDA coating. Only the use of micellar solutions of SDS at this pH provided adequate results in terms of *OE* values (> 90% for the most hydrophobic antidepressants) and NPs reusability (at least for seven cycles). NPs prepared and stored by one year exhibit unaltered efficiency.

The results obtained in this work constitute a starting point for the development of new applications to real samples (i.e., food, physiological and environmental samples) based on the use of surfactants as eluents in D- μ SPE, of particular interest when MS detection is not required or not available.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Mireia Pérez-Baeza: Investigation, Validation, Visualization, Formal analysis, Writing – original draft. **Laura Escuder-Gilabert:** Conceptualization, Methodology, Supervision, Formal analysis, Writing – original draft, Writing – review & editing. **Yolanda Martín-Biosca:** Conceptualization, Methodology, Supervision, Formal analysis, Writing – original draft, Writing – review & editing. **Salvador Sagrado:** Conceptualization, Methodology, Supervision, Formal analysis, Writing – original draft, Writing – review & editing, Software, Data curation. **María José Medina-Hernández:** Writing – review & editing, Project administration, Funding acquisition.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.chroma.2022.463430](https://doi.org/10.1016/j.chroma.2022.463430).

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